No
Date

Untuk dua buah persamaan:
$F(x) = x^2 - x - 2 = 0$ dan $g(x) = x^2 - 4 = 0$, hitunglah pendekatan untuk nilai x sampai iterasi ketiga metode regula falsi
Jawab:

1. $F(x) = x^2 - x - 2 = 0$

Iterasi 1

$a = 1 \quad F(1) = -2$

$b = 3 \quad F(3) = 4$

$$x_1 = \frac{1(4) - 3(-2)}{4 - (-2)} = \frac{4+6}{6} = \frac{10}{6}$$
$$= 1,6667$$

$$F(1,6667) = 1,6667^2 - 1,6667 - 2$$
$$= -0,8889$$

$[a, b] = [1,6667, 3]$

Iterasi 2

$a = 1,6667 \quad F(a) = -0,8889$

$b = 3 \quad F(b) = 4$

$$x_2 = \frac{1,6667(4) - 3(-0,8889)}{4 - (-0,8889)} = \frac{6,6668 + 2,6667}{4,8889}$$
$$= 1,9091$$

$$F(1,9091) = 1,9091^2 - 1,9091 - 2$$
$$= -0,2645$$

$[a, b] = [1,9091, 3]$

Iterasi 3

$a = 1,9091 \quad F(a) = -0,2645$

$b = 3 \quad F(b) = 4$

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$$X_3 = \frac{1{,}9091(4) - 3(-0{,}2645)}{4-(-0{,}2645)} = \frac{7{,}6364 + 0{,}7935}{4{,}2645}$$
$$= 1{,}9767$$

$$X \approx 1{,}9767$$

2. $g(x) = x^2 - 4 = 0$

Iterasi 1

$a = 1 \qquad g(1) = -3$
$b = 3 \qquad g(3) = 5$

$$X_1 = \frac{1(5) - 3(-3)}{5-(-3)} = \frac{5+9}{8}$$
$$= 1{,}7500$$

$$g(1{,}75) = 1{,}75^2 - 4$$
$$= -0{,}9375$$

$$[a,b] = [1{,}75 , 3]$$

Iterasi 2

$a = 1{,}75 \qquad g(a) = -0{,}9375$
$b = 3 \qquad g(b) = 5$

$$X_2 = \frac{1{,}75(5) - 3(-0{,}9375)}{5-(-0{,}9375)} = \frac{8{,}75 + 2{,}8125}{5{,}9375}$$
$$= 1{,}9474$$

$$g(1{,}9474) = 1{,}9474^2 - 4$$
$$= -0{,}2078$$

$$[a,b] = [1{,}9474 , 3]$$

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Iterasi 3

$a = 1,9474$ $\quad g(a) = -0,2078$
$b = 3$ $\quad g(b) = 5$

$$X_3 = \frac{1,9474(5) - 3(-0,2078)}{5 - (-0,2078)}$$
$$= \frac{9,7370 + 0,6234}{5.2078}$$
$$= 1,9895$$

$$X \approx 1,9895$$

$F(x) = x^2 - x - 2$
interval awal = [1, 3]

a	b	F(a)	F(b)	x - r	F(x-r)
1,0000	3,0000	-2,0000	4	1,6667	-0,8889
1,6667	3,0000	-0,8889	4	1,9091	-0,2645
1,9091	3,0000	-0,2645	4	1,9767	-0,0465

$g(x) = x^2 - 4$
interval awal = [1, 3]

a	b	g(a)	g(b)	x - r	g(x-r)
1,0000	3.0000	-3.0000	5	1,7500	-0,9375
1,7500	3.0000	-0,9375	5	1,9474	-0,2078
1,9474	3.0000	-0,2078	5	1,9895	-0,0420

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